5.4.3 Cholesky factorization algorithm (right-looking variant)

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Week 05 due Feb 12, 2025 20:19 IST

Video

from

that we can read off what the operations are that must happen in the

current step and in the future. Namely we know that alpha_11 is equal to lambda_11

squared and we know what alpha_11 is. And therefore we know that lambda_11

must be plus or minus the square root of alpha_11. And we choose to make it equal

to a positive number. And therefore we take it to be equal to the square root

of alpha 11. Now, wait

9:06 / 9:06

2.0x

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Reading assignment

1/1 point (graded)

Read Unit 5.4.3 of the notes. [LINK]

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Typo in Homework 5.4.3.1

According to the lecture and the textbook, shouldn't the update of A22 be: ...

3

Homework 5.4.3.1

1/1 point (graded)

Choose the approximate cost incurred by

$$
\begin{array}{l}
\hline
A = \text{Chol-right-looking}(A) \\
\hline
A \rightarrow \left(\begin{array}{c|c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array} \right) \\
\quad A_{TL} \text{ is } 0 \times 0 \\
\textbf{while } n(A_{TL}) < n(A) \\
\quad \left(\begin{array}{c|c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array} \right) \rightarrow \left(\begin{array}{c|cc} A_{00} & a_{01} & A_{02} \\ \hline a_{10}^T & \alpha_{11} & a_{12}^T \\ A_{20} & a_{21} & A_{22} \end{array} \right) \\
\hline
\quad \alpha_{11} := \lambda_{11} = \sqrt{\alpha_{11}} \\
\quad a_{21} := l_{21} = a_{21} / \alpha_{11} \\
\quad A_{22} := A_{22} - a_{21} a_{21}^H \qquad \text{(syr: update only lower triangular part)} \\
\hline
\quad \left(\begin{array}{c|c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array} \right) \leftarrow \left(\begin{array}{cc|c} A_{00} & a_{01} & A_{02} \\ a_{10}^T & \alpha_{11} & a_{12}^T \\ \hline A_{20} & a_{21} & A_{22} \end{array} \right) \\
\textbf{endwhile} \\
\hline
\end{array}
$$

when applied to an $n \times n$ matrix.

- n^3
- $\frac{2}{3} n^3$
- $\frac{1}{3} n^3$ (selected) ✔

Solution:

For a complete solution, see the notes .

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Answers are displayed within the problem

Homework 5.4.3.2

1/1 point (graded)

Implement the algorithm

$$
\begin{array}{l}
\hline
A = \text{Chol-right-looking}(A) \\
\hline
A \rightarrow \left(\begin{array}{c|c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array} \right) \\
\quad A_{TL} \text{ is } 0 \times 0 \\
\textbf{while } n(A_{TL}) < n(A) \\
\quad \left(\begin{array}{c|c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array} \right) \rightarrow \left(\begin{array}{c|cc} A_{00} & a_{01} & A_{02} \\ \hline a_{10}^T & \alpha_{11} & a_{12}^T \\ A_{20} & a_{21} & A_{22} \end{array} \right) \\
\hline
\quad \alpha_{11} := \lambda_{11} = \sqrt{\alpha_{11}} \\
\quad a_{21} := l_{21} = a_{21} / \alpha_{11} \\
\quad A_{22} := A_{22} - a_{21} a_{12}^T \qquad \text{(syr: update only lower triangular part)} \\
\hline
\quad \left(\begin{array}{c|c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array} \right) \leftarrow \left(\begin{array}{cc|c} A_{00} & a_{01} & A_{02} \\ a_{10}^T & \alpha_{11} & a_{12}^T \\ \hline A_{20} & a_{21} & A_{22} \end{array} \right) \\
\textbf{endwhile} \\
\hline
\end{array}
$$

as

```
function [ A_out ] = Chol_right_looking( A )
```

by completing the code in Assignments/Week05/matlab/Chol_right_looking.m. Input is a HPD $m \times n$ matrix A with only the lower triangular part stored. Output is the matrix A that has its lower triangular part overwritten with the

Cholesky factor. You may want to use Assignments/Week05/matlab/test_Chol_right_looking.m to check your implementation.

- [x] done

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